



THE PROBLEM

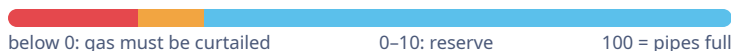
- On the record cold evening of Jan 31, 2024, Enstar told lawmakers it came **"extremely close"** to being unable to deliver gas: storage was maxed out and Hilcorp could add only about **10 million cubic feet** more.¹
- Southcentral runs on **one pressurized system of pipes**. When demand outruns delivery, the gas stored in the pipes drains, pressure falls, and Enstar must cut customers: large users and power plants first, homes last.^{2,3}
- This winter Enstar could start about **3 Bcf short**.⁴
- Losing pressure is slow to undo: in Rhode Island in 2019 it left **7,455 customers** without gas for about a week while every meter was shut off and relit.⁵
- Smart-thermostat programs elsewhere cut gas in short events, but homes reheat afterward. In ConEd's pilot that **snapback** erased about half the saving.⁶

WHAT WE BUILT

A **simulation** of 25,000 Anchorage homes, not control of real thermostats. An operator console shows three charts on one clock: **modeled pipeline pressure**, outdoor temperature (forecast vs actual) and home discomfort.

- Planner**: a linear program over 24 home types and 96 hours finds the least discomfort that keeps pressure above a 10-point reserve every hour, counting the reheating that follows.
- Real forecasts**: it plans with the weather forecasts that existed at the time (archived National Blend of Models runs for Anchorage airport⁷) and **re-plans as each new forecast arrives**, every 6 hours, planning 0.75σ colder than forecast.
- Households**: scan the QR code, join in three taps, see your home on a live map and the system pressure, and override with one tap. The server never holds a home below **62°F**; homes needing steady heat are exempt.

PRESSURE, IN ONE LINE



Each hour the pipes gain what wells and storage can deliver (278 MMcf/day in Feb 2024: the 268 MMcf record day plus ~10 of headroom) and lose what customers burn. Usable pipe storage: 9.54 MMcf. **(derived)** Modeled linepack margin; not psi and not Enstar telemetry.

HOW IT'S BUILT

Browser
React on Vercel;
HiGHS LP in a Web
Worker



SpacetimeDB
runs the house
physics every
second



Every screen
console and
phones update
live; same physics
file on both sides

RESULTS: THE FEB 2024 COLD SNAP, WITH LESS SUPPLY

Nobody was cut off in Feb 2024. We replay that weather and gas demand with less supply than it had. 25,000 homes **(assumed participation)**; numbers **(derived)** from the model, BoreaFlux re-planned on archived forecasts.

Near-miss

11.5 MMcf/day less supply: the most 25,000 homes can cover while holding the full reserve

No program **-7** lowest pressure · 3 h below the line · 0.66 MMcf curtailed

BoreaFlux **+10** lowest pressure · 0 h below · nothing curtailed

Above the curtailment line for the whole cold snap. Cost: about 4.8°F cooler across 22 setback hours (105 °F·h per home); no home below 62°F.

Stress

15 MMcf/day less supply **(assumed case)**

No program **-31** lowest · 17 h below the line · 2.99 MMcf curtailed

BoreaFlux **-16** lowest · 7 h below · 1.57 MMcf curtailed

Curtailment not removed, but about half the gas cut and 10 fewer hours below the line. Cost: 269 °F·h per home.

WHY TRUST THE NUMBERS

Check	Published	Ours	Result
Saving kept after snapback, ConEd-like conditions	48% ⁶	46.5%	Pass (38-58%)
Gas per home on a -20°F day	~1.0 Mcf ⁸	1.04 Mcf	Consistent
Archived forecast error (RMSE), 12 h / 72 h ahead	—	2.7 / 4.7°F	Why we plan colder

Each home's heat loss is calibrated from Enstar's 149 Mcf average home⁸, 30 years of Anchorage airport weather⁹ and Alaska's gas energy content¹⁰. Every number on screen carries a sourced, derived or assumed label.

What it doesn't do. It doesn't close the seasonal 3 Bcf gap or control real thermostats today, and the 0.75σ forecast buffer was tuned on the Feb 2024 replay. It helps on the evenings when pressure decides who gets cut.

SOURCES (EVERY LINK OPENED AND CHECKED, OCT 3-4, 2026)

- Anchorage Daily News, Enstar testimony to the House Energy Committee, Feb 6, 2024. [adn.com/business-economy/energy/2024/02/06/enstar-tells-lawmakers-gas-storage-was-extremely-close-to-being-depleted-during-recent-anchorage-cold-snap/](#)
- Alaska Beacon, Jul 28, 2026 (ADN reprint Jul 29). [adn.com/business-economy/energy/2026/07/29/as-winter-approaches-southcentral-alaska-utilities-are-worried-about-running-short-of-gas/](#)
- ENSTAR tariff \$1220, Interruption Program, eff. Sep 27, 2023. [enstarnaturalgas.com/wp-content/uploads/2023/10/Binder-1200.pdf](#)
- ADN editorial board, Aug 8, 2026. [adn.com/opinions/editorials/2026/08/08/editorial-the-winter-nobody-wants-to-imagine/](#)
- RI Division of Public Utilities and Carriers, Aquidneck Island investigation, Oct 30, 2019. [ripuc.ri.gov/sites/g/files/xkgbur841/files/eventsactions/AI_Report.pdf](#)
- Con Edison, Gas Demand Response pilot report 2019/2020 (NY PSC 17-G-0606). [document.s.dps.ny.gov/public/Common/ViewDoc.aspx?DocRefId=\(7FA119BD-B04D-4378-95A3-BB26697815AC\)](#)
- Iowa Environmental Mesonet, MOS archive (NBS/NBE, PANC). [mesonet.agron.iastate.edu/api/1/docs](#)
- ADN, Energy Watch drill results, Nov 6, 2012. [adn.com/energy/article/results-are-latest-southcentral-alaska-energy-watch-drill/2012/11/06/](#)
- ACIS, Anchorage airport daily data 1996-2025. [rcc-acis.org/docs_webservices.html](#)
- U.S. EIA, heat content of delivered gas, Alaska, 2025. [eia.gov/dnav/ng/ng_cons_heat_a_epg0_vgth_btucf_a.htm](#)